Lecture 3 — Statistical Inference

Introductory Econometrics — Télécom Paris

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Reading: Wooldridge (5th ed.) sections 4.1–4.6

Recap: what Lecture 2 tells us about OLS

Under MLR.1–MLR.5: a linear model, random sampling, no perfect collinearity, zero conditional mean and homoskedasticity.

$$y = X\beta + u, \qquad \hat{\beta} = \beta + (X'X)^{-1}X'u.$$

X contains the sample values of all regressors, including the intercept.

$$\mathbb{E}(\hat{\beta} \mid X) = \beta, \qquad \mathrm{Var}(\hat{\beta} \mid X) = \sigma^2(X'X)^{-1}.$$

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Estimate the variance and standard deviation using

$$\hat{\sigma}^2 = \frac{\sum_i \hat{u}_i^2}{n-k-1}, \qquad \widehat{\mathrm{Var}}(\hat{\beta} \mid X) = \hat{\sigma}^2(X'X)^{-1}, \qquad \mathrm{se}(\hat{\beta}_j) = \hat{\sigma}\sqrt{[(X'X)^{-1}]_{jj}}.$$

The mean describes the estimator's center; the variance describes its precision. For statistical inference, we need the full sampling distribution of the estimators.

Inference about population regression parameters

We now use sample estimates to test hypotheses about the parameters in the population model.

1. Find the OLS sampling distributions under normally distributed errors.
2. Test a hypothesis about one population parameter.
3. Test a single hypothesis involving several parameters.
4. Test several restrictions jointly.

What can a sample tell us about the education coefficient?

$$wage_i = \beta_0 + \beta_1 educ_i + \beta_2 exper_i + \beta_3 tenure_i + u_i.$$

The estimated education coefficient is \$0.599 per hour per additional year, holding experience and tenure fixed. Another sample would give another estimate.

Is that estimate compatible with a population coefficient $\beta_1 = 0$? Or with $\beta_1 = 0.50$?

Inference uses the sampling variation of $\hat{\beta}_1$ to assess claims about β_1.

Probability review: the normal distribution

If $Z \sim N(0,1)$, its density is

$$\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}, \qquad \Pr(a < Z \leq b) = \int_a^b \phi(z)\, dz.$$

For $W \sim N(\mu, \sigma^2)$, with $\sigma > 0$,

$$\mathbb{E}(W) = \mu, \qquad \mathrm{Var}(W) = \sigma^2, \qquad \frac{W-\mu}{\sigma} \sim N(0,1).$$

A normal distribution is characterized by two parameters: the mean and the variance.

The q quantile z_q satisfies $\Pr(Z \leq z_q) = q$ for $Z \sim N(0,1)$.

q	0.95	0.975	0.99	0.995
z_q	1.645	1.960	2.326	2.576

Probability review: jointly normal variables

A vector $Z \in \mathbb{R}^m$ is jointly normal if and only if, for all $a \in \mathbb{R}^m$, $a'Z$ is normal.

A standard normal vector has mean zero and covariance matrix I_m:

$$Z \sim N_m(0, I_m).$$

Each component is $N(0,1)$, and the components are independent. For jointly normal variables, zero covariance implies independence.

A multivariate normal distribution is characterized by its mean vector $\mu \in \mathbb{R}^m$ and its covariance matrix $\Sigma \in \mathbb{R}^{m \times m}$:

$$Z \sim N_m(\mu, \Sigma).$$

Probability review: standardizing a normal vector

For a symmetric ($\Sigma = \Sigma'$), positive definite matrix Σ of dimension $m \times m$, there exists a symmetric matrix $\Sigma^{-1/2}$ such that

$$\Sigma^{-1/2}\Sigma\Sigma^{-1/2} = I_m.$$

With this positive definite Σ, the following distributional statements are equivalent:

$$Z \sim N_m(\mu, \Sigma) \quad \iff \quad \Sigma^{-1/2}(Z - \mu) \sim N_m(0, I_m).$$

Corollary. If $Z \sim N_m(\mu, \Sigma)$ and $A \in \mathbb{R}^{q \times m}$ is nonrandom,

$$\underbrace{AZ}_{q \times 1} \sim N_q\left(A\mu, \underbrace{A\Sigma A'}_{q \times q}\right).$$

Probability review: the chi-square distribution

For independent $Z_1, \ldots, Z_m \sim N(0,1)$,

$$Q = Z_1^2 + \cdots + Z_m^2 = Z'Z \sim \chi^2_m, \qquad \mathbb{E}(Q) = m, \quad \mathrm{Var}(Q) = 2m.$$

m is the number of degrees of freedom: the number of independent squared standard normal components.

For $W \sim N_m(\mu, \Sigma)$ with positive definite Σ,

$$(W-\mu)'\Sigma^{-1}(W-\mu) \sim \chi^2_m.$$

This will help us describe the distribution of sums of squared regression errors.

Sampling Distributions of $\hat{\beta}^{\mathrm{OLS}}$: parametric model

To obtain a tractable sampling distribution for $\hat{\beta}^{\mathrm{OLS}}$, we assume:

Assumption MLR.6 Normality: The population error u is independent of the explanatory variables $\mathrm{x} = (1, x_1, \ldots, x_k)'$ and is normally distributed with mean zero and variance σ^2: $u \mid \mathrm{x} \sim N(0, \sigma^2)$.

Together with the linear population model, MLR.6 implies

$$y \mid \mathrm{x} \sim N(\mathrm{x}'\beta, \sigma^2).$$

MLR.2 gives independent draws of (y_i, x_i). Conditional on the full sample matrix X, the errors are therefore independent:

$$u = (u_1, \ldots, u_n)', \qquad u \mid X \sim N_n(0, \sigma^2 I_n).$$

Sampling Distributions of $\hat{\beta}^{\mathrm{OLS}}$: classical linear model

MLR.6 is stronger than the previous error assumptions: $\mathbb{E}(u \mid \mathrm{x}) = 0$, $\mathrm{Var}(u \mid \mathrm{x}) = \sigma^2$.

In cross-sectional regression, MLR.1–MLR.6 define the **classical linear model (CLM)**.

Under MLR.1–MLR.6, the OLS estimators $\hat{\beta}^{\mathrm{OLS}}$ are the minimum variance unbiased estimators.

Sampling Distributions of $\hat{\beta}^{\mathrm{OLS}}$

Condition on the sample values of the regressors, X: $\hat{\beta} = \beta + (X'X)^{-1}X'u$.

- Once X is held fixed, the errors are the only source of randomness.
- $\hat{\beta}$ is an affine combination of normal random variables (MLR.1–MLR.6).
- With: $\mathbb{E}(\hat{\beta} \mid X) = \beta$, $\mathrm{Var}(\hat{\beta} \mid X) = \sigma^2(X'X)^{-1}$.

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- With: $\mathbb{E}(\hat{\beta} \mid X) = \beta$, $\mathrm{Var}(\hat{\beta} \mid X) = \sigma^2(X'X)^{-1}$.

Therefore, the joint distribution is

$$\hat{\beta} \mid X \sim N_{k+1}\big(\beta, \sigma^2(X'X)^{-1}\big)\,.$$

Thus,

$$\left.\frac{\hat{\beta}_j - \beta_j}{\mathrm{sd}(\hat{\beta}_j \mid X)}\right| X \sim N(0,1), \qquad j = 0, \ldots, k.$$

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The true standard deviation depends on the unknown error standard deviation σ.

Testing hypotheses about a single population parameter:

The t test

Testing a population coefficient

We can test $H_0 : \beta_j = c$ using the t statistic of $\hat{\beta}_j$:

$$t_{\hat{\beta}_j}(c) \equiv \frac{\hat{\beta}_j - c}{\hat{\sigma}\sqrt{[(X'X)^{-1}]_{jj}}}.$$

Under MLR.1–MLR.6 and H_0, $t_{\hat{\beta}_j}(c) \mid X \sim t_{n-k-1}$.

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- H_0: Once $\boldsymbol{x}_{-j}$ have been accounted for, is x_j's effect on $\mathbb{E}(y \mid x)$ equal to c?
- $t_{\hat{\beta}_j}(c)$ measures how many estimated standard deviations $\hat{\beta}_j$ is away from c.
- Intuition: suppose $\beta_j = c$. How unusual would our observed standardized distance be under repeated sampling?

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- Intuition: suppose $\beta_j = c$. How unusual would our observed standardized distance be under repeated sampling?
- A sample estimate far from c, relative to its standard error, provides evidence against H_0.
- The precise rejection rule depends on the alternative and the chosen significance level.

Deriving the distribution: A feasible test

We want to test $H_0 : \beta_j = c$: for example, is the education coefficient zero, or equal to \$0.50?

Under H_0, the standardized distance from the proposed value is:

$$\left| \frac{\hat{\beta}_j - c}{\sigma \sqrt{[(X'X)^{-1}]_{jj}}} \right| \; X \sim N(0,1).$$

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$$\left. \frac{\hat{\beta}_j - c}{\sigma\sqrt{[(X'X)^{-1}]_{jj}}} \right| X \sim N(0,1).$$

We cannot compute it because σ^2 is unknown. Estimate it using the residuals:

$$\hat{\sigma}^2 = \frac{\sum_i \hat{u}_i^2}{n-k-1}.$$

We can then define the test statistic using the estimated standard deviation:

$$T_j(c) = \frac{\hat{\beta}_j - c}{\hat{\sigma}\sqrt{[(X'X)^{-1}]_{jj}}}.$$

To judge whether this distance is unusually large, we need its distribution under H_0.

Deriving the distribution: Residual projections

Work under MLR.1–MLR.6, conditional on X, with $n > k + 1$ and $\sigma^2 > 0$.

To study $\hat{\sigma}^2 = \hat{u}'\hat{u}/(n-k-1)$, express the residuals in terms of the population errors:

$$\begin{aligned}\hat{u} &= y - X\hat{\beta} \\ &= X\beta + u - X\left[\beta + (X'X)^{-1}X'u\right] \\ &= \left[I_n - X(X'X)^{-1}X'\right]u = Mu.\end{aligned}$$

Define the fitted-value and residual matrices:

$$P = X(X'X)^{-1}X', \qquad M = I_n - P.$$

P projects orthogonally onto the column space of X: $Py = \hat{y}$.

M projects orthogonally onto its orthogonal complement: $My = \hat{u}$.

Deriving the distribution: The residual distribution

The residual matrix is symmetric and idempotent:

$$M' = M, \qquad M^2 = (I_n - P)^2 = I_n - P = M,$$

because $P' = P$ and $P^2 = X(X'X)^{-1}X'X(X'X)^{-1}X' = P$.

Since $\hat{u} = Mu$ and $u \mid X \sim N_n(0, \sigma^2 I_n)$,

$$\mathbb{E}(\hat{u} \mid X) = 0, \qquad \mathrm{Var}(\hat{u} \mid X) = \sigma^2 MM' = \sigma^2 M, \qquad \frac{\hat{u}}{\sigma} \,\middle|\, X \sim N_n(0, M).$$

The residual vector satisfies $k+1$ independent linear restrictions, $X'\hat{u} = 0$, and thus has $\nu = n - k - 1$ degrees of freedom.

The raw residuals are correlated in general and do not all have variance σ^2, so their standardized sum of squares is not immediately a sum of independent $N(0,1)$ squares.

Changing coordinates will let us write $\hat{u}'\hat{u}/\sigma^2$ as a sum of $n - k - 1$ squared independent $N(0,1)$ variables.

Deriving the distribution: Orthonormal coordinates

Conditional on X, choose the orthonormal basis

$$U = \begin{pmatrix} U_X & U_\perp \end{pmatrix}, \qquad U'U = UU' = I_n.$$

U_X contains $k+1$ directions spanning the column space of X; $U_\perp$ contains $\nu = n-k-1$ directions spanning its orthogonal complement.

U is a coordinate-change matrix. For any vector v, $U'v$ gives its coordinates in this basis:

$$v = U(U'v), \qquad (U'v)'(U'v) = v'v.$$

Applying M to these directions gives $MU_X = 0$ and $MU_\perp = U_\perp$. Therefore,

$$U'MU = \begin{pmatrix} 0_{k+1} & 0 \\ 0 & I_\nu \end{pmatrix}.$$

The coordinates change; the vector and its squared length stay the same.

Deriving the distribution: The residual sum of squares

Rotate the standardized errors: $z = U'u/\sigma$, so $z \mid X \sim N_n(0, I_n)$.

Then express the standardized residuals in this basis, using $\hat{u} = Mu$ and inserting $UU' = I_n$:

$$\frac{U'\hat{u}}{\sigma} = U'MU\frac{U'u}{\sigma} = \begin{pmatrix} 0_{(k+1)\times 1} \\ (z_{k+2}, \ldots, z_n)' \end{pmatrix}.$$

Because $UU' = I_n$, the rotation preserves the sum of squares:

$$\frac{\hat{u}'\hat{u}}{\sigma^2} = \left(\frac{U'\hat{u}}{\sigma}\right)' \left(\frac{U'\hat{u}}{\sigma}\right) = \sum_{\ell=k+2}^{n} z_\ell^2.$$

These are $\nu = n - k - 1$ independent squared standard normals. Therefore

$$\frac{\nu\hat{\sigma}^2}{\sigma^2} = \frac{\hat{u}'\hat{u}}{\sigma^2}, \qquad \left.\frac{\hat{u}'\hat{u}}{\sigma^2}\right| X \sim \chi^2_\nu.$$

The same $\hat{u}'\hat{u}/\sigma^2$ is therefore a sum of $n - k - 1$ squared independent $N(0, 1)$ variables.

Deriving the distribution: Independence

We know the numerator's normal law and the variance estimator's chi-square law. We still need to establish their independence.

Conditional on X, $\hat{\beta} = \beta + (X'X)^{-1}X'u$ and $\hat{u} = Mu$ are jointly normal: both are affine transformations of the same normal vector u.

$$\begin{aligned}\mathrm{Cov}(\hat{\beta}, \hat{u} \mid X) &= (X'X)^{-1}X'\mathrm{Var}(u \mid X)M' \\ &= \sigma^2(X'X)^{-1}X'M \\ &= 0,\end{aligned}$$

since $X'M = X' - X'X(X'X)^{-1}X' = 0$.

Joint normality turns this zero covariance into independence:

$$\hat{\beta} \perp\!\!\!\perp \hat{u} \mid X \quad \Longrightarrow \quad \hat{\beta} \perp\!\!\!\perp \hat{\sigma}^2 \mid X, \qquad \hat{\sigma}^2 = \frac{\hat{u}'\hat{u}}{\nu}.$$

Independence is preserved when we take a function of the residual vector.

Deriving the distribution: Student's t distribution

Definition: Student's t distribution. Let

$$Z \sim N(0,1), \qquad Q \sim \chi^2_\nu, \qquad Z \perp\!\!\!\perp Q.$$

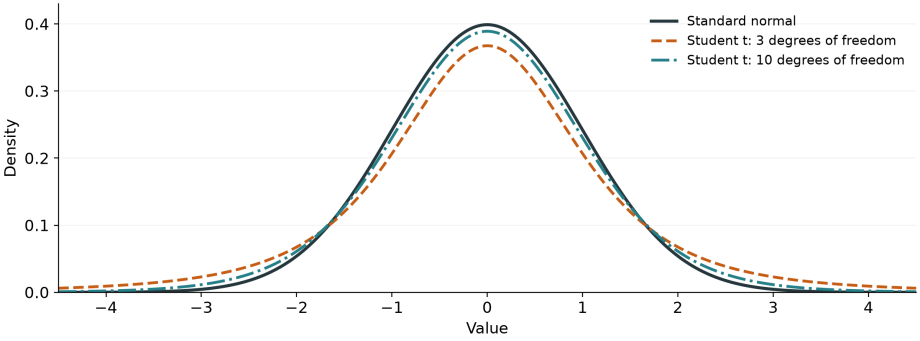
Then

$$T = \frac{Z}{\sqrt{Q/\nu}} \sim t_\nu.$$

The denominator is random: $\mathbb{E}(Q/\nu) = 1$ and $\mathrm{Var}(Q/\nu) = 2/\nu$. Small realizations of the denominator can produce large absolute values of T.

This is the form we need for a coefficient divided by its estimated standard error.

Deriving the distribution: Extra uncertainty



The distribution is symmetric about zero and has heavier tails than $N(0,1)$. As the degrees of freedom increase, the curves approach the standard normal density.

Deriving the distribution: The estimated standard error

For coefficient j, the true and estimated standard deviations are

$$\mathrm{sd}(\hat{\beta}_j \mid X) = \sigma\sqrt{[(X'X)^{-1}]_{jj}}, \qquad \mathrm{se}(\hat{\beta}_j) = \hat{\sigma}\sqrt{[(X'X)^{-1}]_{jj}}.$$

Conditional on X, the common matrix factor is fixed. With $\nu = n - k - 1$,

$$\frac{\hat{\sigma}\sqrt{[(X'X)^{-1}]_{jj}}}{\sigma\sqrt{[(X'X)^{-1}]_{jj}}} = \frac{\hat{\sigma}}{\sigma} = \sqrt{\frac{Q}{\nu}}, \qquad Q = \frac{\nu\hat{\sigma}^2}{\sigma^2}.$$

We have shown that $Q \mid X \sim \chi^2_\nu$ and that Q is independent of $\hat{\beta}_j$ conditional on X.

These are precisely the denominator and independence required by the definition of t.

Deriving the distribution: The studentized coefficient

Under MLR.1–MLR.6, conditional on X, define

$$Z_j = \frac{\hat{\beta}_j - \beta_j}{\sigma\sqrt{[(X'X)^{-1}]_{jj}}} \sim N(0,1), \qquad Q = \frac{\nu\hat{\sigma}^2}{\sigma^2} \sim \chi^2_\nu, \qquad Z_j \perp\!\!\!\perp Q.$$

Dividing the numerator and denominator by the true standard deviation gives

$$\frac{\hat{\beta}_j - \beta_j}{\hat{\sigma}\sqrt{[(X'X)^{-1}]_{jj}}} = \frac{\dfrac{\hat{\beta}_j - \beta_j}{\sigma\sqrt{[(X'X)^{-1}]_{jj}}}}{\hat{\sigma}/\sigma} = \frac{Z_j}{\sqrt{Q/\nu}} \sim t_\nu.$$

Under $H_0 : \beta_j = c$, the test statistic is

$$T_j(c) = \frac{\hat{\beta}_j - c}{\hat{\sigma}\sqrt{[(X'X)^{-1}]_{jj}}}, \qquad T_j(c) \mid X \sim t_{n-k-1}.$$

This gives the reference distribution for the test statistic we can compute.

Deriving the distribution: The normal limit

For independent standard normals $W_1, W_2, \ldots,$

$$Q_\nu = \sum_{\ell=1}^{\nu} W_\ell^2 \sim \chi^2_\nu, \qquad \frac{Q_\nu}{\nu} = \frac{1}{\nu}\sum_{\ell=1}^{\nu} W_\ell^2 \xrightarrow{p} \mathbb{E}(W_\ell^2) = 1.$$

The law of large numbers makes the random scale increasingly close to one. For an independent $Z \sim N(0,1)$,

$$T_\nu = \frac{Z}{\sqrt{Q_\nu/\nu}} \xrightarrow{d} N(0,1) \qquad \text{as } \nu \to \infty.$$

With few residual degrees of freedom, the extra uncertainty matters more. Under the normal model, t_ν remains the exact finite-sample distribution.

Testing against one-sided alternatives

One-sided alternatives: hypotheses and evidence

Under MLR.1–MLR.6 and the hypothesis $H_0 : \beta_j = c$,

$$t_{\hat{\beta}_j} = \frac{\hat{\beta}_j - c}{\hat{\sigma}\sqrt{[(X'X)^{-1}]_{jj}}}, \qquad t_{\hat{\beta}_j} \mid X \sim t_{n-k-1}.$$

- Recall t_{n-k-1} is centered around zero.
- If $H_0 : \beta_j \leq c$ or $H_0 : \beta_j \geq c$, we test at the boundary $H_0 : \beta_j = c$, the hardest to reject.

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To determine a rule for rejecting H_0, we need to decide on the relevant alternative hypothesis.

- Under $H_1 : \beta_j > c$, we expect $t_{\hat{\beta}_j} > 0 \rightarrow$ we look for a sufficiently **positive** value of $t_{\hat{\beta}_j}$.
- Under $H_1 : \beta_j < c$, we expect $t_{\hat{\beta}_j} < 0 \rightarrow$ we look for a sufficiently **negative** value of $t_{\hat{\beta}_j}$.

One-sided alternatives: rejection rules

$H_0 : \beta_j = 0$ is rejected in favor of $H_1 : \beta_j > 0$ at the α significance level if $t_{\hat{\beta}_j} > t_{n-k-1,\,1-\alpha}$.

- The **significance level** α is the probability of rejecting $H_0 : \beta_j = 0$ when it is in fact true.
- $t_{n-k-1,\,1-\alpha}$ is the $(1-\alpha)$ quantile of the t_{n-k-1} distribution.

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- $t_{n-k-1,\,1-\alpha}$ is the $(1-\alpha)$ quantile of the t_{n-k-1} distribution.
- The critical value satisfies $\Pr_{H_0}\left(t_{\hat{\beta}_j} > t_{n-k-1,\,1-\alpha} = a \mid X\right) = \alpha$.
- Rejecting $H_0 : \beta_j = 0$ in favor of $H_1 : \beta_j > 0$ also rejects the broader null $H_0 : \beta_j \leq 0$ (subtracting a negative proposed value makes the t statistic larger).

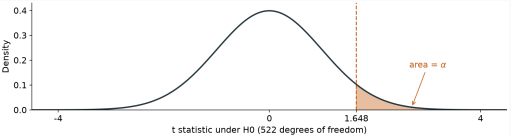
For $H_0 : \beta_j = c$ against $H_1 : \beta_j < c$, reject if $t_{\hat{\beta}_j} < -a$. By symmetry, $\Pr_{H_0}\left(t_{\hat{\beta}_j} < -a \mid X\right) = \alpha$.

One-sided alternatives: rejection regions

For $H_0 : \beta_j = c$, take $\alpha = 0.05$ and $n - k - 1 = 522$, so $a = 1.648$.

Right-sided alternative: $H_1 : \beta_j > c$.

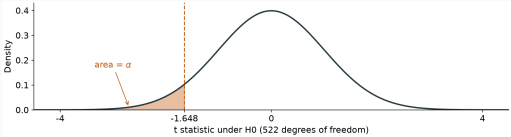
Reject if $t_{\hat{\beta}_j} > a = 1.648$.



A negative statistic provides no evidence for this right-sided alternative.

Left-sided alternative: $H_1 : \beta_j < c$.

Reject if $t_{\hat{\beta}_j} < -a = -1.648$.



Only a sufficiently negative statistic leads to rejection in favor of this alternative.

Each shaded region has probability α at the null boundary.

One-sided alternatives: an experience coefficient

Set $c = 0$ and test $H_0 : \beta_{exper} = 0$ against $H_1 : \beta_{exper} > 0$. Let $n = 526$ and $n - k - 1 = 522$.

$$\widehat{\log(wage)} = \underset{(0.104)}{0.284} + \underset{(0.007)}{0.092}\, educ + \underset{(0.0017)}{0.0041}\, exper + \underset{(0.003)}{0.022}\, tenure.$$

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Set $c = 0$ and test $H_0 : \beta_{exper} = 0$ against $H_1 : \beta_{exper} > 0$. Let $n = 526$ and $n - k - 1 = 522$.

$$\widehat{\log(wage)} = \underset{(0.104)}{0.284} + \underset{(0.007)}{0.092}\, educ + \underset{(0.0017)}{0.0041}\, exper + \underset{(0.003)}{0.022}\, tenure.$$

Since $t_{\hat{\beta}_{exper}} \approx 2.41 > t_{522,\,0.99} \approx 2.334$, reject H_0 at 1% against the positive alternative.

$\hat{\beta}_{exper}$ is statistically greater than zero at the 1% significance level.

The reported t statistic is calculated from the rounded estimates: $0.0041/0.0017 \approx 2.41$.

Testing against two-sided alternatives

Two-sided alternatives: hypotheses and test statistic

Does the population coefficient equal c, or does it differ in either direction?

$$H_0 : \beta_j = c \qquad \text{against} \qquad H_1 : \beta_j \neq c.$$

Under MLR.1–MLR.6, the statistic and its distribution under H_0 are

$$t_{\hat{\beta}_j} = \frac{\hat{\beta}_j - c}{\hat{\sigma}\sqrt{[(X'X)^{-1}]_{jj}}}, \qquad t_{\hat{\beta}_j} \mid X \sim t_{n-k-1}.$$

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- A sufficiently positive or negative statistic provides evidence against H_0: the test uses the **absolute value** $|t_{\hat{\beta}_j}|$.
- For $c = 0$, this is the usual two-sided **test of statistical significance**: is the coefficient on x_j zero, holding the other regressors fixed?

Two-sided alternatives: rejection rule

Choose the significance level α before observing the test statistic. For $H_0 : \beta_j = c$ against $H_1 : \beta_j \neq c$, the critical value satisfies

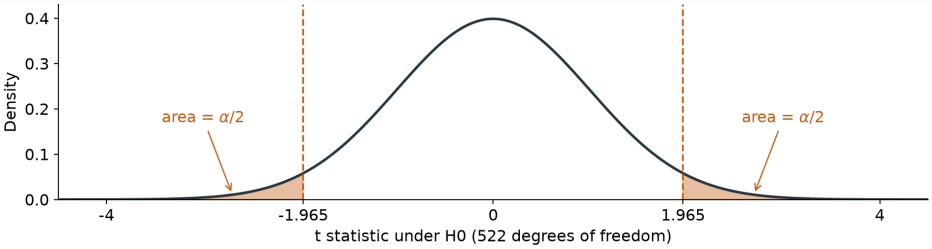
$$\Pr_{H_0}\Big(|t_{\hat{\beta}_j}| > a \mid X\Big) = \alpha, \qquad a = t_{n-k-1,\,1-\alpha/2}.$$

- a is the $(1 - \alpha/2)$ quantile of the t_{n-k-1} distribution.
- **Reject** H_0 at level α if $|t_{\hat{\beta}_j}| > a$: equivalently, if $t_{\hat{\beta}_j} < -a$ or $t_{\hat{\beta}_j} > a$.
- **Fail to reject** H_0 if $|t_{\hat{\beta}_j}| \leq a$. This does not establish that $\beta_j = c$.

For $c = 0$, rejection means that $\hat{\beta}_j$ is **statistically significant** (different from zero), at the α significance level.

Two-sided alternatives: rejection regions

Under $H_0 : \beta_j = c$, $t_{\hat{\beta}_j} \mid X \sim t_{n-k-1}$. Take $\alpha = 0.05$ and $n - k - 1 = 522$, so $a = t_{522,\,0.975} \approx 1.965$.



Reject H_0	**Fail to reject H_0**	**Reject H_0**
$t_{\hat{\beta}_j} < -1.965$	$-1.965 \leq t_{\hat{\beta}_j} \leq 1.965$	$t_{\hat{\beta}_j} > 1.965$

The two rejection regions together have probability $0.025 + 0.025 = 0.05$ under H_0.

Computing p-values for t Tests

A critical-value test gives a decision at a chosen significance level.

The **p-value** is the probability, under H_0, of a test statistic at least as extreme as the one observed, t_{obs}, in the direction specified by H_1.

Computing p-values for t Tests

A critical-value test gives a decision at a chosen significance level.

The **p-value** is the probability, under H_0, of a test statistic at least as extreme as the one observed, t_{obs}, in the direction specified by H_1.

- It gives the smallest significance level at which H_0 would be rejected given the observed value of the t statistic.
- It allows us to compare the evidence with any significance level.
- **Decision rule:** reject H_0 at level α if $p < \alpha$; otherwise, fail to reject. The p-value is the threshold level at which the decision changes.

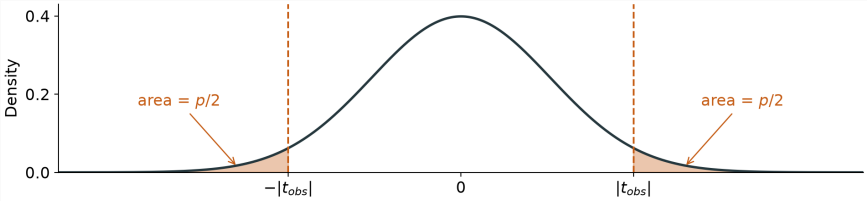
Computing p-values for t Tests

Two-sided alternative: $H_0 : \beta_j = c$ against $H_1 : \beta_j \neq c$.

For $T \sim t_{n-k-1}$, symmetry gives

$$p = \Pr(|T| \geq |t_{obs}|) = \Pr(T \leq -|t_{obs}|) + \Pr(T \geq |t_{obs}|) = 2\Pr(T \geq |t_{obs}|).$$

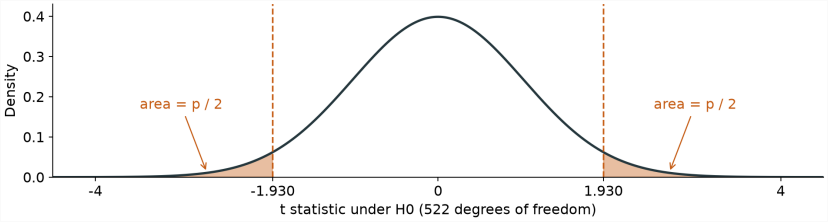
Let $F_{n-k-1}(z) = \Pr(T \leq z)$ be the CDF. Then $p = 2\,[1 - F_{n-k-1}(|t_{obs}|)]$.



Computing p-values for t Tests

In the wage example, test $H_0 : \beta_{educ} = 0.50$ against $H_1 : \beta_{educ} \neq 0.50$.

$$t_{obs} = \frac{\hat{\beta}_{educ} - 0.50}{\text{se}(\hat{\beta}_{educ})} \approx 1.930, \qquad T \sim t_{522}, \qquad p = 2\,\Pr(T \geq 1.930) \approx 0.054.$$



Under H_0, an absolute t statistic at least this large occurs with probability about 5.4%.

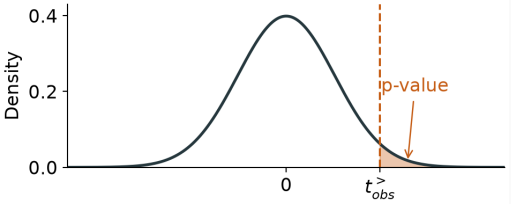
Since $0.05 < p < 0.10$, fail to reject at 5%, but reject at 10%.

Computing p-values for t Tests

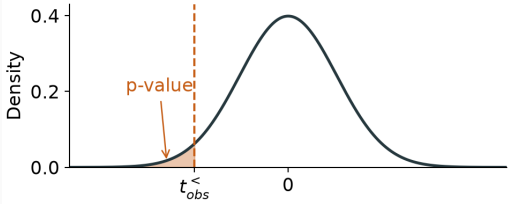
One-sided alternatives: use the tail specified by H_1 and the signed value t_{obs}. For $H_0 : \beta_j = c$, take $T \sim t_{n-k-1}$ and write $F_{n-k-1}(z) = \Pr(T \le z)$.

Alternative	Observed t	Tail probability	Computation
$H_1 : \beta_j > c$	$t^{>}_{obs}$	$p = \Pr(T \ge t^{>}_{obs})$	$1 - F_{n-k-1}(t^{>}_{obs})$
$H_1 : \beta_j < c$	$t^{<}_{obs}$	$p = \Pr(T \le t^{<}_{obs})$	$F_{n-k-1}(t^{<}_{obs})$

$H_1 : \beta_j > c$



$H_1 : \beta_j < c$



The language of hypothesis testing

When we cannot reject $H_0 : \beta_x = c$, we say that we failed to reject H_0 at the $x\%$ level.

Rather than: "H_0 is accepted at the $x\%$ level."

We might be unable to reject H_0, but there are many other values for β_x that cannot be rejected.

Therefore, it makes no sense to say we accept either of these hypotheses, as they cannot be true at the same time.

Statistical significance and economic importance

Example 4.6: Regress 401(k) participation (*prate*, in percent) on the employer match rate (*mrate*), plan age (*age*) and number of employees (*totemp*):

$$\widehat{prate} = \underset{(0.78)}{80.29} + \underset{(0.52)}{5.44}\, mrate + \underset{(0.045)}{0.269}\, age - \underset{(0.00004)}{0.00013}\, totemp, \qquad n = 1{,}534, \quad R^2 = 0.100.$$

The smallest t statistic in absolute value is on *totemp*: $t_{\hat{\beta}_{totemp}} = -0.00013/0.00004 = -3.25$.

- Its two-sided p-value is about 0.001, thus *totemp* is statistically significant at 1%.
- **Economic importance** concerns the magnitude and units. Holding *mrate* and *age* fixed, 10,000 additional employees correspond to a fitted participation change of -1.3 percentage points: a modest change for a very large increase in firm size.

Confidence intervals

Confidence intervals

Under MLR.1–MLR.6, the standardized estimation error satisfies

$$\frac{\hat{\beta}_j - \beta_j}{\hat{\sigma}\sqrt{[(X'X)^{-1}]_{jj}}} \,\Bigg|\, X \sim t_{n-k-1}.$$

Let $a = t_{n-k-1,\,1-\alpha/2}$. Then

$$\Pr\left(-a \le \frac{\hat{\beta}_j - \beta_j}{\hat{\sigma}\sqrt{[(X'X)^{-1}]_{jj}}} \le a \,\middle|\, X\right) = 1 - \alpha.$$

$$\Pr\left(\hat{\beta}_j - a\hat{\sigma}\sqrt{[(X'X)^{-1}]_{jj}} \le \beta_j \le \hat{\beta}_j + a\hat{\sigma}\sqrt{[(X'X)^{-1}]_{jj}} \,\middle|\, X\right) = 1 - \alpha.$$

Thus, a $(1-\alpha)$ confidence interval for β_j is

$$\left[\hat{\beta}_j - a\hat{\sigma}\sqrt{[(X'X)^{-1}]_{jj}},\quad \hat{\beta}_j + a\hat{\sigma}\sqrt{[(X'X)^{-1}]_{jj}}\right].$$

Confidence intervals: what does 95% mean?

Imagine repeatedly drawing samples and constructing an interval by the same rule. Under the model, 95% of those intervals contain the fixed population coefficient.

The interval's endpoints vary across samples; β_j does not. For the interval computed from our observed sample, the coefficient is either inside or outside.

For a fixed confidence level, a smaller estimated standard deviation gives a narrower interval. For the same sample, raising the confidence level requires a larger critical value and a wider interval.

This is an interval for a population coefficient, not an interval containing 95% of individual outcomes. Its coverage relies on the inference assumptions.

Testing a single linear combination of the parameters

A single linear combination: hypothesis and distribution

For a fixed, nonzero vector $\boldsymbol{\lambda} \in \mathbb{R}^{k+1}$, test

$$H_0 : \boldsymbol{\lambda}'\boldsymbol{\beta} = \gamma \qquad \text{against} \qquad H_1 : \boldsymbol{\lambda}'\boldsymbol{\beta} \neq \gamma.$$

Under MLR.1–MLR.6, a linear combination of the jointly normal OLS estimators is normal:

$$\boldsymbol{\lambda}'\hat{\boldsymbol{\beta}} \mid X \sim N\big(\boldsymbol{\lambda}'\boldsymbol{\beta},\, \sigma^2 \boldsymbol{\lambda}'(X'X)^{-1}\boldsymbol{\lambda}\big)\,.$$

Example: constant returns to scale. With output Y, labor L and capital K,

$$\log Y = \beta_0 + \beta_1 \log L + \beta_2 \log K + u.$$

Constant returns requires $\beta_1 + \beta_2 = 1$:

$$\boldsymbol{\lambda} = (0, 1, 1)', \qquad \boldsymbol{\lambda}'\boldsymbol{\beta} = \beta_1 + \beta_2, \qquad H_0 : \boldsymbol{\lambda}'\boldsymbol{\beta} = 1.$$

A single linear combination: t statistic and rejection rule

Under MLR.1–MLR.6 and $H_0 : \boldsymbol{\lambda}'\boldsymbol{\beta} = \gamma$, conditional on X,

$$Z = \frac{\boldsymbol{\lambda}'\hat{\boldsymbol{\beta}} - \gamma}{\sigma\sqrt{\boldsymbol{\lambda}'(X'X)^{-1}\boldsymbol{\lambda}}} \sim N(0,1), \qquad Q = \frac{(n-k-1)\hat{\sigma}^2}{\sigma^2} \sim \chi^2_{n-k-1},$$

and Z and Q are independent. Therefore,

$$t = \frac{\boldsymbol{\lambda}'\hat{\boldsymbol{\beta}} - \gamma}{\hat{\sigma}\sqrt{\boldsymbol{\lambda}'(X'X)^{-1}\boldsymbol{\lambda}}} = \frac{Z}{\sqrt{Q/(n-k-1)}}, \qquad t \mid X \sim t_{n-k-1}.$$

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For $H_1 : \boldsymbol{\lambda}'\boldsymbol{\beta} \neq \gamma$, reject H_0 at significance level α if

$$|t| > a = t_{n-k-1,\,1-\alpha/2}, \qquad \Pr_{H_0}(|t| > a \mid X) = \alpha.$$

There is **one restriction**, even when several coefficients enter $\boldsymbol{\lambda}'\boldsymbol{\beta}$. The denominator uses the standard error of the combination, **including covariances**.

Testing multiple linear restrictions:

The F test

Multiple linear restrictions: the joint hypothesis

Objective: test several linear restrictions on the population coefficients simultaneously.

Example: Let N_1, N_2 be nominal values and I_1, I_2 their price indices. Consider

$$Y = \beta_0 + \beta_1 \log N_1 + \beta_2 \log I_1 + \beta_3 \log N_2 + \beta_4 \log I_2 + u.$$

Since N_j/I_j is the real quantity and $\log N_j = \log(N_j/I_j) + \log I_j$, rewrite the same model as

$$Y = \beta_0 + \beta_1 \log \frac{N_1}{I_1} + (\beta_1 + \beta_2) \log I_1 + \beta_3 \log \frac{N_2}{I_2} + (\beta_3 + \beta_4) \log I_2 + u.$$

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We want to test whether **only the real quantities matter**, so both extra price terms disappear:

$$H_0 : \beta_1 + \beta_2 = 0 \text{ and } \beta_3 + \beta_4 = 0.$$

The alternative is that at least one sum is nonzero.

Multiple linear restrictions: matrix representation

Write $\boldsymbol{\beta} = (\beta_0, \beta_1, \beta_2, \beta_3, \beta_4)'$. The two restrictions $\beta_1 + \beta_2 = 0$ and $\beta_3 + \beta_4 = 0$ can be written using

$$R = \begin{pmatrix} 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 \end{pmatrix}, \qquad \boldsymbol{\gamma} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}.$$

Thus,

$$H_0 : R\boldsymbol{\beta} = \boldsymbol{\gamma} \qquad \text{against} \qquad H_1 : R\boldsymbol{\beta} \neq \boldsymbol{\gamma}.$$

- Each row of R defines one linear combination; the corresponding entry of $\boldsymbol{\gamma}$ gives its hypothesized value.
- The alternative means that **at least one restriction fails**. We need one statistic that accounts for the joint sampling uncertainty of all the combinations.

The F distribution: definition

Let U_1 and U_2 be independent random variables with

$$U_1 \sim \chi^2_{d_1}, \qquad U_2 \sim \chi^2_{d_2}.$$

Then the ratio of the two chi-square variables, each divided by its degrees of freedom,

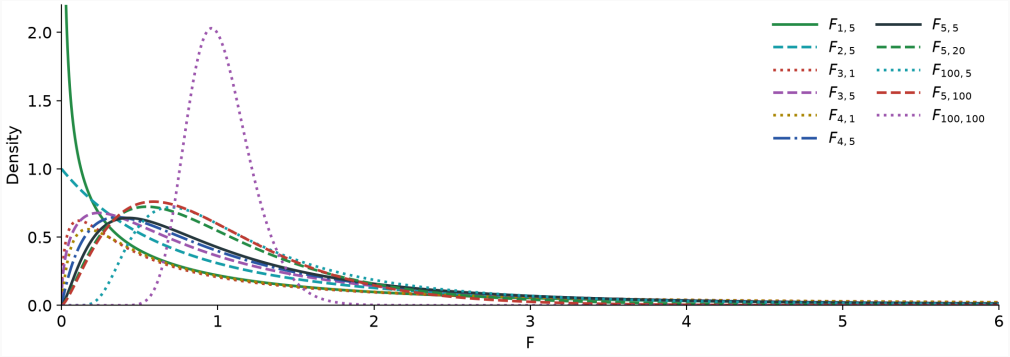
$$F = \frac{U_1/d_1}{U_2/d_2} \sim F_{d_1, d_2},$$

has an **F distribution** (Fisher distribution).

- d_1 is the numerator degrees of freedom; d_2 is the denominator degrees of freedom.
- The statistic is nonnegative. The order of the two degrees of freedom matters.
- In our regression test, these will be q restrictions and $n - k - 1$ residual degrees of freedom.

The F distribution: density curves

The shape depends on both the numerator and denominator degrees of freedom.



With both degrees of freedom large, the distribution becomes concentrated near one.

Deriving the test and its distribution: rank and normality

The general null hypothesis is

$$H_0 : \underbrace{R}_{q\times(k+1)} \underbrace{\boldsymbol{\beta}}_{(k+1)\times 1} = \underbrace{\boldsymbol{\gamma}}_{q\times 1}.$$

Assume R is fixed and has **full row rank** q: its restrictions are linearly independent. A redundant row adds no restriction and should not be counted again.

Under MLR.1–MLR.6,

$$\hat{\boldsymbol{\beta}} \mid X \sim N_{k+1}\big(\boldsymbol{\beta}, \sigma^2 (X'X)^{-1}\big),$$

so the vector of estimated linear combinations satisfies

$$R\hat{\boldsymbol{\beta}} \mid X \sim N_q\big(R\boldsymbol{\beta}, \sigma^2 R(X'X)^{-1}R'\big).$$

Deriving the test and its distribution: a chi-square numerator

Under MLR.1–MLR.6 and $H_0 : R\boldsymbol{\beta} = \boldsymbol{\gamma}$,

$$R\hat{\boldsymbol{\beta}} - \boldsymbol{\gamma} \mid X \sim N_q\big(0, \sigma^2 R(X'X)^{-1}R'\big)\,.$$

Standardizing the vector gives

$$\boldsymbol{Z} = \frac{1}{\sigma}\left[R(X'X)^{-1}R'\right]^{-1/2}\left(R\hat{\boldsymbol{\beta}} - \boldsymbol{\gamma}\right), \qquad \boldsymbol{Z} \mid X \sim N_q(0, I_q).$$

For $C = R(X'X)^{-1}R'$, the inverse square root is the symmetric positive-definite matrix $C^{-1/2}$ satisfying $(C^{-1/2})^2 = C^{-1}$.

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For $C = R(X'X)^{-1}R'$, the inverse square root is the symmetric positive-definite matrix $C^{-1/2}$ satisfying $(C^{-1/2})^2 = C^{-1}$.

We need one measure of joint departure from H_0: summing the q squared independent standard normal entries prevents positive and negative deviations from cancelling and gives a known null distribution:

$$U_1 = \boldsymbol{Z}'\boldsymbol{Z} = \frac{(R\hat{\boldsymbol{\beta}} - \boldsymbol{\gamma})'\left[R(X'X)^{-1}R'\right]^{-1}(R\hat{\boldsymbol{\beta}} - \boldsymbol{\gamma})}{\sigma^2}, \qquad U_1 \mid X \sim \chi^2_q.$$

Deriving the test and its distribution: the F statistic

Under MLR.1–MLR.6, the residual variance estimate satisfies

$$U_2 = \frac{(n-k-1)\hat{\sigma}^2}{\sigma^2}, \qquad U_2 \mid X \sim \chi^2_{n-k-1}.$$

Conditional on X, U_1 and U_2 are independent: the first is a function of $\hat{\boldsymbol{\beta}}$, which is independent of $\hat{\sigma}^2$ under the normal model.

Therefore, under $H_0 : R\boldsymbol{\beta} = \boldsymbol{\gamma}$,

$$F = \frac{U_1/q}{U_2/(n-k-1)} = \frac{(R\hat{\boldsymbol{\beta}} - \boldsymbol{\gamma})' \left[R(X'X)^{-1}R'\right]^{-1} (R\hat{\boldsymbol{\beta}} - \boldsymbol{\gamma})}{q\hat{\sigma}^2},$$

$$F \mid X \sim F_{q,n-k-1}.$$

The unknown σ^2 cancels. A large statistic indicates a large joint departure from the restrictions, relative to sampling uncertainty.

Application: joint significance and decision

For **joint significance**, R selects q coefficients and $H_0 : R\boldsymbol{\beta} = 0$. The alternative is that at least one selected coefficient is nonzero.

$$\text{All coefficients: } R = I_{k+1}, \qquad \text{Only } \beta_0, \beta_1 : \; R = \begin{pmatrix} 1 & 0 & 0 & \cdots & 0 \\ 0 & 1 & 0 & \cdots & 0 \end{pmatrix}.$$

Under MLR.1–MLR.6 and H_0,

$$F = \frac{\hat{\boldsymbol{\beta}}' R' \left[R(X'X)^{-1} R' \right]^{-1} R\hat{\boldsymbol{\beta}}}{q\hat{\sigma}^2}, \qquad F \mid X \sim F_{q,n-k-1}.$$

The critical value satisfies $a = F_{q,n-k-1,\,1-\alpha}$, $\Pr_{H_0}(F > a \mid X) = \alpha$.

Application: joint significance and decision

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The critical value satisfies $a = F_{q,n-k-1,\,1-\alpha}$, $\Pr_{H_0}(F > a \mid X) = \alpha$.

- **Reject** H_0 at level α if $F_{obs} > a$; otherwise, fail to reject.
- The rejection region is in the upper tail.
- Joint significance does not imply that every coefficient is individually significant.

Multiple linear restrictions: two coefficients jointly (Example)

Test $H_0 : \beta_1 = \beta_2 = 0$ against $H_1 : \beta_1 \neq 0$ or $\beta_2 \neq 0$. For $\boldsymbol{\beta} = (\beta_0, \beta_1, \beta_2, \ldots, \beta_k)'$,

$$R = \begin{pmatrix} 0 & 1 & 0 & 0 & \cdots & 0 \\ 0 & 0 & 1 & 0 & \cdots & 0 \end{pmatrix}, \qquad R\hat{\boldsymbol{\beta}} = \begin{pmatrix} \hat{\beta}_1 \\ \hat{\beta}_2 \end{pmatrix}.$$

Write the symmetric inverse matrix as

$$\left[R(X'X)^{-1}R'\right]^{-1} = \begin{pmatrix} b_{11} & b_{12} \\ b_{12} & b_{22} \end{pmatrix}.$$

Then the joint statistic is

$$F = \frac{b_{11}\hat{\beta}_1^2 + 2b_{12}\hat{\beta}_1\hat{\beta}_2 + b_{22}\hat{\beta}_2^2}{2\hat{\sigma}^2} \quad \sim F_{2,n-k-1} \quad \text{under } H_0, \text{ conditional on } X.$$

Reject if $F_{obs} > F_{2,n-k-1,\,1-\alpha}$. The cross term reflects the joint covariance structure; this is one test of two restrictions.

Testing multiple linear restrictions:

The Fisher test through restricted least squares

The Fisher test: from Wald to restricted least squares

So far, we have tested $H_0 : R\boldsymbol{\beta} = \boldsymbol{\gamma}$ against $H_1 : R\boldsymbol{\beta} \neq \boldsymbol{\gamma}$ using the **Wald approach**.

It measures how far the unrestricted estimate $R\hat{\boldsymbol{\beta}}$ is from $\boldsymbol{\gamma}$, relative to its estimated covariance matrix:

$$F = \frac{(R\hat{\boldsymbol{\beta}} - \boldsymbol{\gamma})' \left[R(X'X)^{-1}R'\right]^{-1} (R\hat{\boldsymbol{\beta}} - \boldsymbol{\gamma})}{q\hat{\sigma}^2} \quad \Big| \; X \sim F_{q,n-k-1}$$

under MLR.1–MLR.6 and H_0, with q independent restrictions.

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under MLR.1–MLR.6 and H_0, with q independent restrictions.

Alternative construction: impose H_0 when estimating the model and measure how much the restrictions worsen its fit. This gives the **Fisher test** in terms of residual sums of squares.

In this linear model, the two constructions give **exactly the same F statistic**.

The Fisher test: statistic and rejection rule

Let SSR_r and SSR_{ur} be the restricted and unrestricted residual sums of squares. For q independent restrictions $H_0 : R\boldsymbol{\beta} = \boldsymbol{\gamma}$, under MLR.1–MLR.6 and H_0,

$$F = \frac{(SSR_r - SSR_{ur})/q}{SSR_{ur}/(n-k-1)}, \qquad F \mid X \sim F_{q,n-k-1}.$$

This equals the Wald statistic (proof omitted); imposing restrictions gives $SSR_r \geq SSR_{ur}$.

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This equals the Wald statistic (proof omitted); imposing restrictions gives $SSR_r \geq SSR_{ur}$.

- Estimate both models on the **same observations**. $k+1$ is the number of coefficients in the unrestricted model.
- The numerator is the loss of fit per restriction; the denominator is $\hat{\sigma}^2$ from the unrestricted model.
- A sufficiently large loss of fit, relative to estimated noise, is evidence against H_0.

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- The numerator is the loss of fit per restriction; the denominator is $\hat{\sigma}^2$ from the unrestricted model.
- A sufficiently large loss of fit, relative to estimated noise, is evidence against H_0.

Reject H_0 at level α if $F_{obs} > a = F_{q,n-k-1,1-\alpha}$, where $\Pr_{H_0}(F > a \mid X) = \alpha$.

The Fisher test: exclusion restrictions

Partition the full model as

$$y = X_1\beta_1 + X_2\beta_2 + u, \qquad H_0 : \beta_2 = 0,$$

where X_2 contains q regressors and X_1 contains the remaining $k + 1 - q$ columns.

- **Restricted model:** regress y on X_1 alone.

$$\hat{u}_r = y - X_1\hat{\beta}_{1,r}, \qquad SSR_r = \hat{u}_r'\hat{u}_r.$$

- **Unrestricted model:** regress y on both X_1 and X_2.

$$\hat{u} = y - X_1\hat{\beta}_1 - X_2\hat{\beta}_2, \qquad SSR_{ur} = \hat{u}'\hat{u}.$$

Compute $F = \dfrac{(SSR_r - SSR_{ur})/q}{SSR_{ur}/(n-k-1)}$. The retained coefficients are re-estimated in the restricted regression.

The Fisher test: wage example

Test $H_0 : \beta_{exper} = \beta_{tenure} = 0$ against the alternative that at least one is nonzero.

	Restricted	Unrestricted
Dependent variable	*wage*	*wage*
Regressors	1, *educ*	1, *educ*, *exper*, *tenure*
Observations	526	526
Residual sum of squares	5980.68	4966.30

There are $q = 2$ restrictions and $n - k - 1 = 522$ unrestricted residual degrees of freedom:

$$F_{obs} = \frac{(5980.68 - 4966.30)/2}{4966.30/522} \approx 53.310.$$

Since $53.310 > F_{2,522,\,0.95} \approx 3.013$, reject H_0 at 5%: experience and tenure are **jointly significant**, controlling for education.

The Fisher test: p-value

For an observed statistic F_{obs}, the **p-value** is the probability under H_0 of a statistic at least as large. Under MLR.1–MLR.6,

$$p = \Pr_{H_0}(F \geq F_{obs} \mid X) = 1 - G_{q,n-k-1}(F_{obs}),$$

where $G_{q,n-k-1}$ is the CDF of the $F_{q,n-k-1}$ distribution.

- **Decision rule:** reject H_0 if $p < \alpha$; otherwise, fail to reject.
- For exclusion restrictions, rejection means that the excluded coefficients are jointly significant. It does not establish that each is individually significant.
- Failure to reject does not prove the restrictions or justify automatically dropping the variables.

In the wage example, $F_{obs} \approx 53.310$ and $p < 0.001$.

The Fisher test: relationship with the t test

For one restriction $H_0 : \lambda'\beta = \gamma$, $q = 1$ and

$$t = \frac{\lambda'\hat{\beta} - \gamma}{\hat{\sigma}\sqrt{\lambda'(X'X)^{-1}\lambda}}, \qquad F = \frac{(\lambda'\hat{\beta} - \gamma)^2}{\hat{\sigma}^2\lambda'(X'X)^{-1}\lambda} = t^2.$$

Under MLR.1–MLR.6 and H_0,

$$t \mid X \sim t_{n-k-1} \quad \Longrightarrow \quad t^2 \mid X \sim F_{1,n-k-1}.$$

The critical values satisfy

$$F_{1,n-k-1,\,1-\alpha} = \left(t_{n-k-1,\,1-\alpha/2}\right)^2.$$

Thus, the F test and the **two-sided t test** have the same decision and p-value. Squaring removes the sign, so it does not reproduce a one-sided test.

The Fisher test: the R-squared form

For exclusion restrictions, use the same dependent variable and observations, with an intercept in both regressions. Both models have the same $SST = \sum_i (y_i - \bar{y})^2$:

$$R^2_{ur} = 1 - \frac{SSR_{ur}}{SST}, \qquad R^2_r = 1 - \frac{SSR_r}{SST}.$$

Substitute $SSR = SST(1 - R^2)$ into the Fisher statistic:

$$\begin{aligned} F &= \frac{\{SST(1 - R^2_r) - SST(1 - R^2_{ur})\}/q}{SST(1 - R^2_{ur})/(n - k - 1)} \\ &= \frac{(R^2_{ur} - R^2_r)/q}{(1 - R^2_{ur})/(n - k - 1)}. \end{aligned}$$

This is the same test expressed as the gain in R^2 when the restrictions are removed. Use ordinary R^2, not adjusted R^2.

The Fisher test: overall significance

The overall-significance test reported by regression software tests all slopes jointly:

$$H_0 : \beta_1 = \cdots = \beta_k = 0 \quad \text{against} \quad H_1 : \text{at least one slope is nonzero.}$$

The restricted model is $y_i = \beta_0 + u_i$, so $\hat{y}_{i,r} = \bar{y}$, $SSR_r = SST$, $R_r^2 = 0$ and $q = k$. The intercept remains unrestricted.

Under MLR.1–MLR.6 and H_0,

$$F = \frac{R^2/k}{(1 - R^2)/(n - k - 1)}, \qquad F \mid X \sim F_{k,n-k-1}.$$

Reject at level α if $F_{obs} > F_{k,n-k-1,\,1-\alpha}$.

In the wage model, $F_{obs} = 76.873$ with $(3, 522)$ degrees of freedom. Overall significance does not establish a large R^2, correct specification or causality.

Exact inference in the normal linear model

Under MLR.1–MLR.6, conditioning on the sample values of the regressors gives exact sampling distributions for $\hat{\boldsymbol{\beta}}$ and $\hat{\sigma}^2$.

- **One linear restriction:** a t statistic with $n - k - 1$ degrees of freedom. The same distribution yields confidence intervals.
- q **independent linear restrictions:** an F statistic with $(q, n - k - 1)$ degrees of freedom.
- **Two equivalent constructions:** the Wald approach uses the distance from the restrictions; the Fisher approach uses the loss of fit from imposing them.

Lecture 4 asks how to conduct inference when normality is unavailable and we rely on large-sample approximations.

Reporting regression results

Reporting results: estimates and standard errors together

Dependent variable: hourly wage	(1)	(2)	(3)
Education (years)	0.541	0.644	0.599
	(0.053)	(0.054)	(0.051)
Experience (years)	—	0.070	0.022
		(0.011)	(0.012)
Tenure (years)	—	—	0.169
			(0.022)
Intercept	-0.905	-3.391	-2.873
	(0.685)	(0.767)	(0.729)
Observations	526	526	526
R^2	0.165	0.225	0.306

Conventional standard errors in parentheses; these columns use the same 526 workers.

Reporting results: make the estimates usable

Identify the dependent variable and its units, the explanatory variables in each specification, the sample size and R^2.

Report standard errors with estimates so readers can assess precision, form confidence intervals and test values other than zero.

For a test central to the research question, state the null and alternative, report the p-value or rejection rule, and explain the conclusion in the context of the question.

Distinguish the estimated magnitude, its precision and its economic interpretation. Statistical significance alone does not summarize the result.